

Train a Regression Model

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Absolute Error and MAE

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Regression and Residuals

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Score validation for Fitting a line. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Fitting a line. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Training (0, 2),(1, 5),(2, 8). Candidates $m_2 b_2$, $m_3 b_1$, $m_3 b_2$. Select by SSE, then score (3, 10),(4, 15) with MAE.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Training points $(x,y)=(1,3),(2,5),(3,7)$. Compare $y=x+1$ and $y=2x+1$ by training MAE; select one, then predict held-out $x=4$, actual=10. Explain what was fitted.

6. Solve the given case

Supplied fitted rule: height=2 times age+1. Held-out observations are (3,8),(4,9).

Predict and evaluate; distinguish inference.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A regression model has validation MAE 2.4 cm; a training-mean baseline has validation MAE 2.0 cm on the same examples. Which performs better on this metric?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 22 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Add the regression experiment to the project appendix only as a separate, clearly limited model.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Training points $(x,y)=(1,3),(2,5),(3,7)$. Compare $y=x+1$ and $y=2x+1$ by training MAE; select one, then predict held-out $x=4$, actual=10. Explain what was fitted.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Trace candidates

Give the three training SSE values and selected line.

My reasoning and evidence

13. W2 Score validation

Predict at x 3 and x 4 and calculate MAE/MSE.

My reasoning and evidence

14. W3 Compare the baseline

Fit the training constant and score it on validation.

My reasoning and evidence

15. W4 Apply independently

Training (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10),(4, 15) with MAE.

My reasoning and evidence

16. W5 Limit extrapolation

Why is an exact training line not proof of correct predictions at $x = 100$?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Train a Regression Model

Fit a toy growth line from training pairs, then leave evaluation labels out of fitting.

Training(day, height):(0, 1),(1, 3),(2, 5). Candidates $y=2x+1$ and $y=x+1$. Evaluation pair:(3, 6).

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Train a Regression Model

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Calculate training fit

$2x+1$ predicts 1, 3, 5 with SSE0; $x+1$ predicts 1, 2, 3 with SSE5.



2. Select line

Choose $2x+1$ under minimum training SSE.



3. Evaluate once

At day3 predict 7 versus observed 6: absolute error 1 cm.

CHANGE ONE THING

Evaluation observation becomes 8 instead of 6.

Explain the changed result using the matching lesson.

May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Fit a toy growth line from training pairs, then leave evaluation labels out of fitting.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? At day3 predict7 versus observed6: absolute error1 cm.

1. Select line
2. Calculate training fit
3. Evaluate once

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Evaluate once At day3 predict7 versus observed6: absolute error1 cm. Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Evaluation observation becomes8 instead of6. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Absolute evaluation error stays 1 cm, but residual changes from -1 to +1. Prediction 7 remains frozen; changed observation 8 reverses the error direction.

Repair a missing building block

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as 4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric prediction from classification. Short practice: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
